CAREFLOW

AI Powered Intelligent Hospital Patient Flow Management System

FINAL REVIEW · COMPLETE PROJECT

A. RAMU · 23K91A6715 B. RUSHI · 23K91A6720
B. RAJ KUMAR · 23K91A6725 A. PRABHAS · 23K91A6703

Guide: [GUIDE NAME] · CSE (Data Science) · Batch A10

Evidence-based documentation · Synthetic model data · Real running-system captures

NAVIGATION

Contents

01 Problem and research

02 Objectives and approach

03 Architecture and methodology

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Abstract

- CareFlow coordinates patient visits across queues, consultation, laboratory and bed state.
- One persisted Visit record drives role dashboards and a public board.
- Random Forest estimates wait; Gradient Boosting estimates triage priority.
- Evidence is synthetic-model evaluation plus local software verification.

7 Departments

13 Captured screens

Base Paper

- Alleviate, IEEE Access, 2025.
- Integrated mobile healthcare access, recommendation, prescriptions, maps and planning.
- Its 97.24% precision concerns specialist recommendation.
- CareFlow changes the problem to internal operational flow; metrics are not comparable.

97.24% Different task

2025 IEEE Access

Literature Survey

- Access and facility-location systems.
- Disease and specialist recommendation.
- Personalised diet/exercise and monitoring.
- Healthcare chatbots and NLP.
- Waiting-time prediction and interpretability.

19 References

4 Survey themes

Research Gap

- No single shared state across tokens, movement, tests and beds.
- Prediction studies often stop at a score, without operational action.
- Patient-facing apps do not coordinate the internal hospital journey.
- Evidence boundaries between synthetic metrics and real impact are often unclear.

Gap — **Operational integration**

Need — **Auditable state**

Problem Statement

- Coordinate patients through OPD, emergency, laboratory, pharmacy, ward and ICU while preserving priority and current state.
- Give each role the actions and visibility needed for its part of the journey.
- Use AI assistance without overstating clinical validity or operational impact.

One Visit state

Many Role views

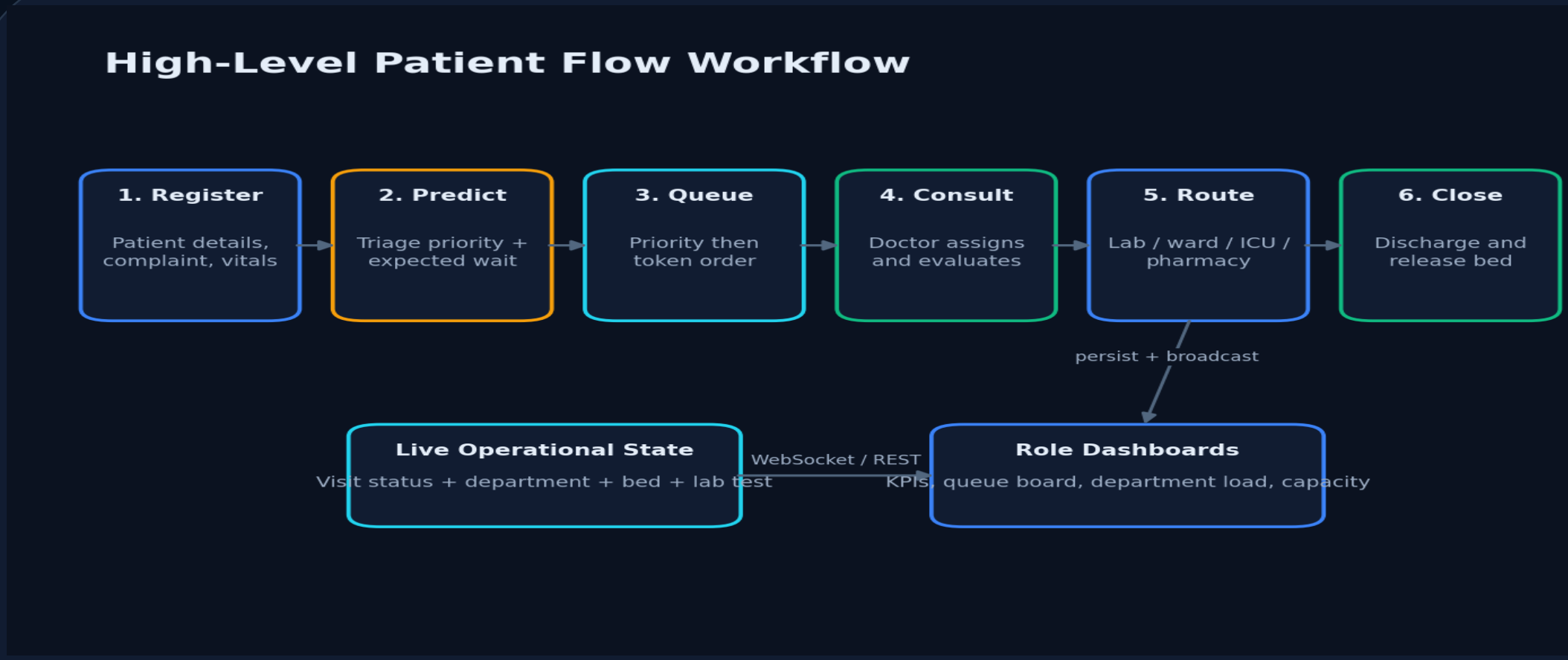
Objectives

- Digital tokens and priority queues.
- Visit state and movement tracking.
- Laboratory workflow.
- Live ward and ICU capacity.
- Triage and wait prediction.
- Role dashboards and public board.

6 Project objectives

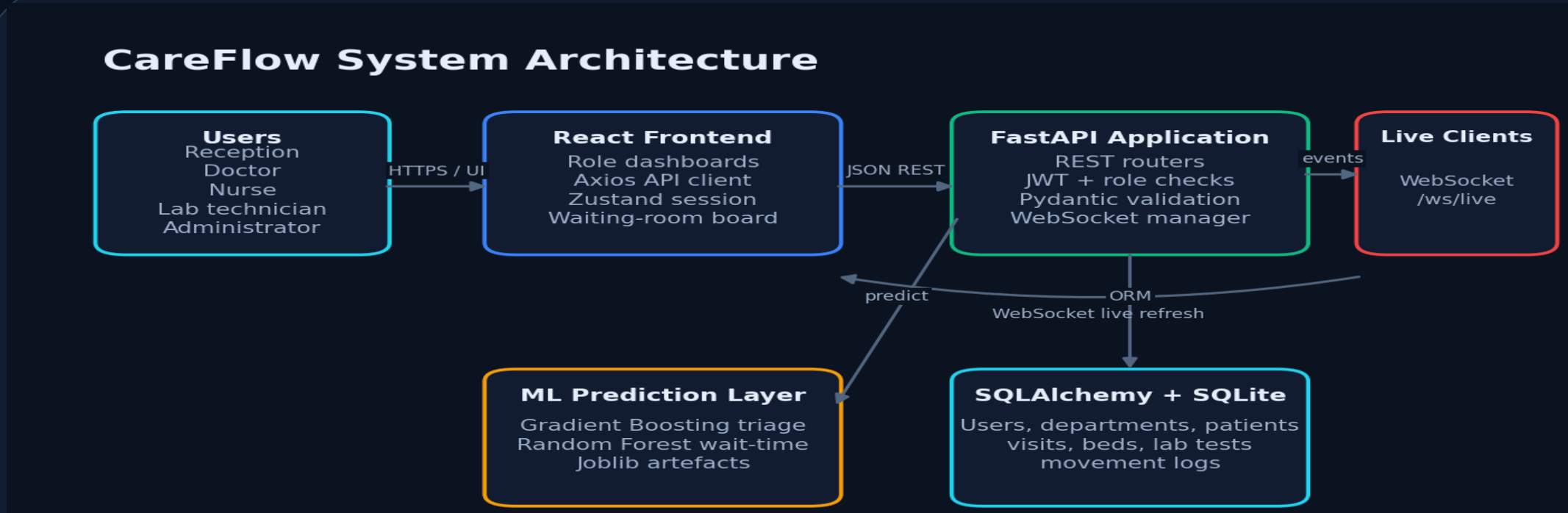
5+1 Views

Proposed System



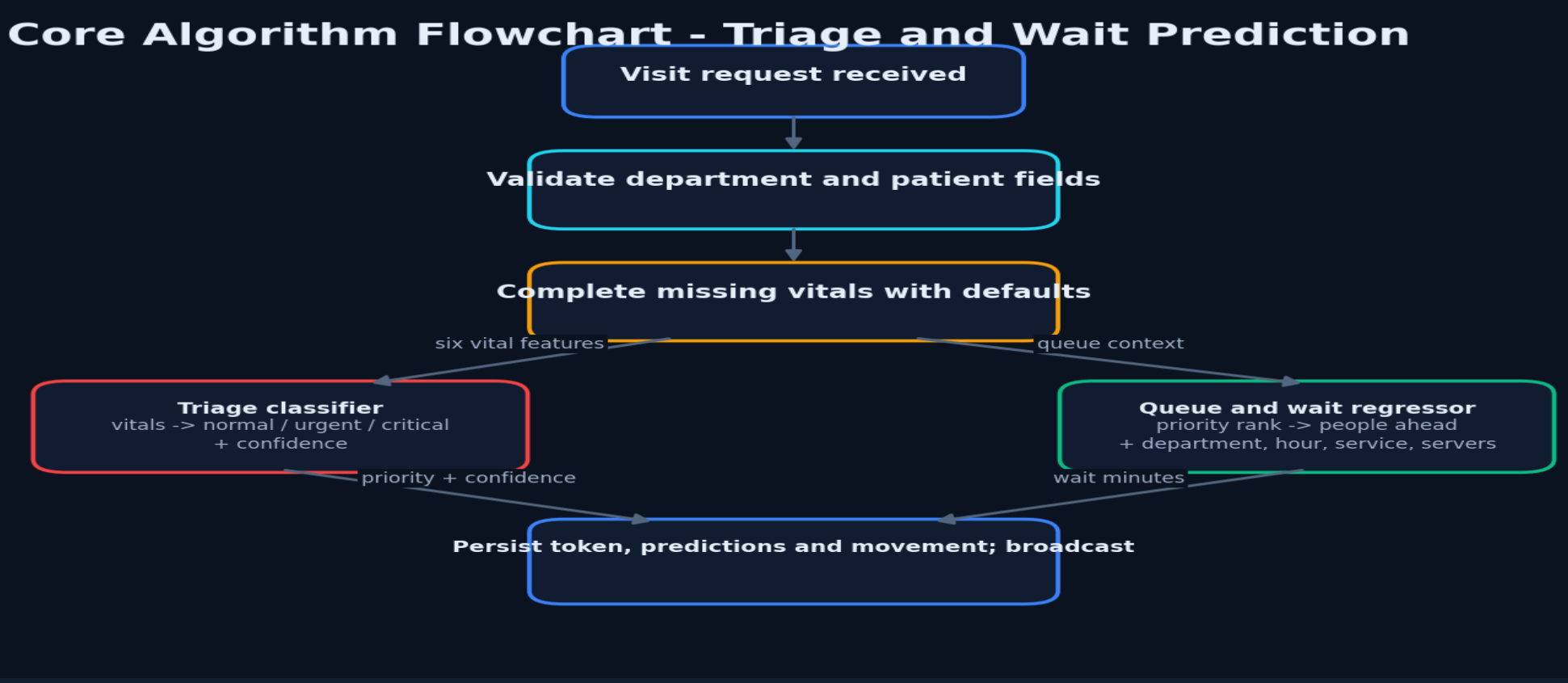
End-to-end operational flow

System Architecture



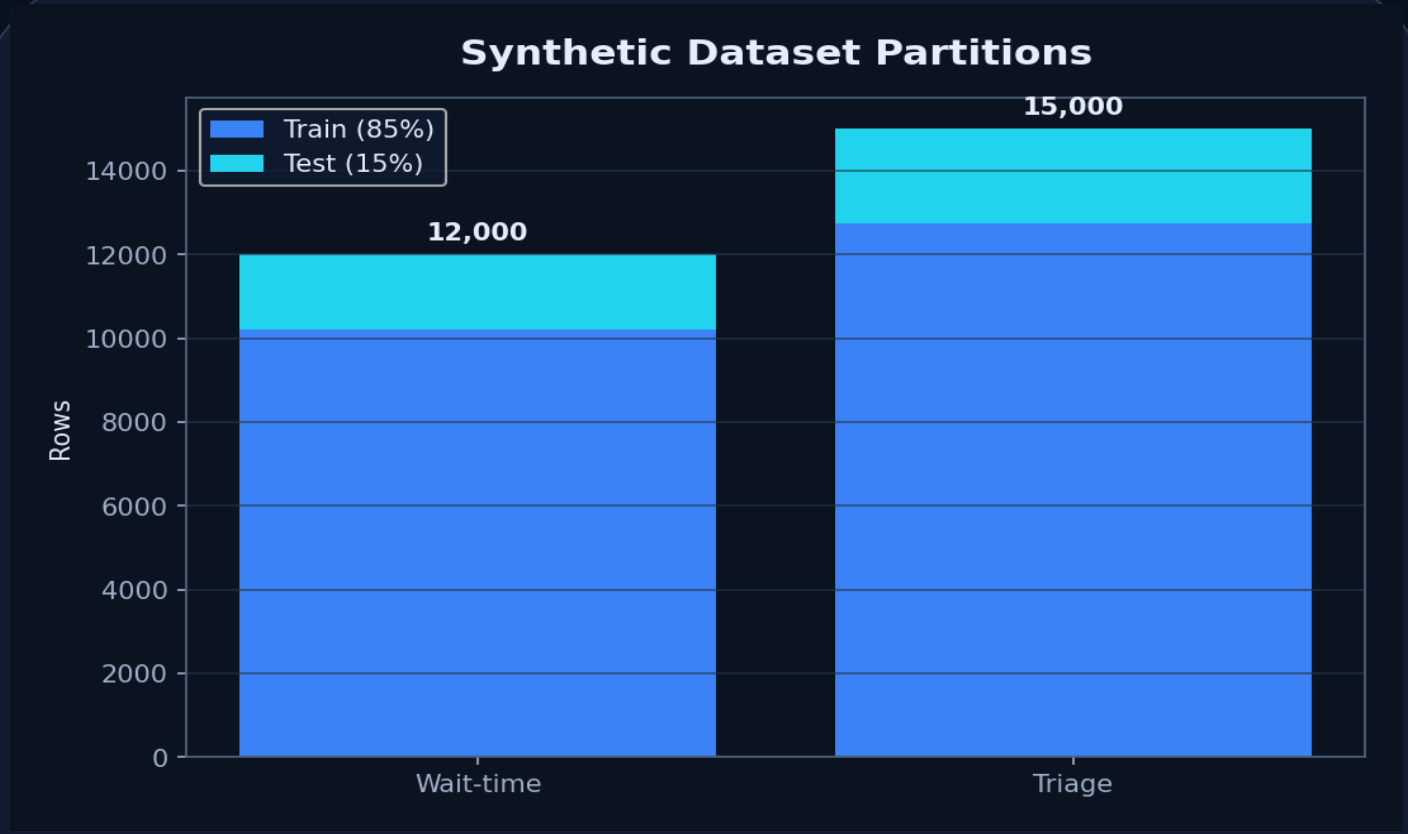
React, FastAPI, ML, SQLite and WebSocket

Methodology



Triage and wait inference integrated with persistence

Dataset and Models



Synthetic datasets and 85/15 splits

- RF: 200 trees
- GB: 150 estimators
- Seed 42
- No real patient data

Decision Logic

- Missing vital components are completed with documented defaults.
- Classifier returns normal, urgent or critical plus confidence.
- Queue-ahead count respects priority rank.
- Regressor returns non-negative wait minutes.
- Visit and movement are committed before broadcast.

3 Priority classes

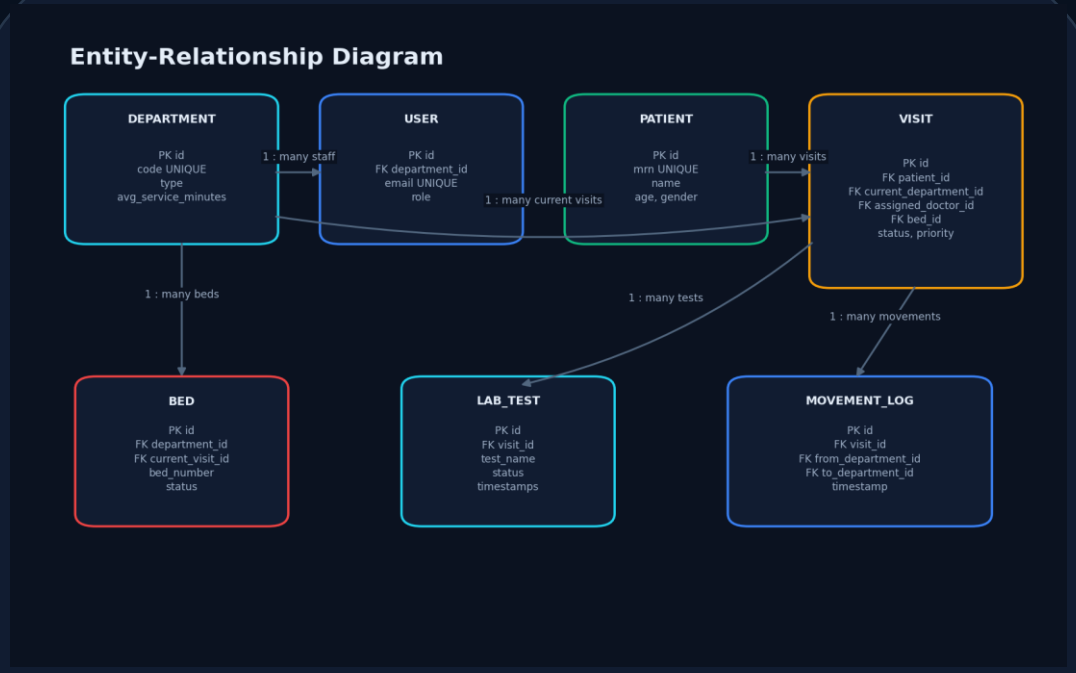
2 Default servers

Data Flow

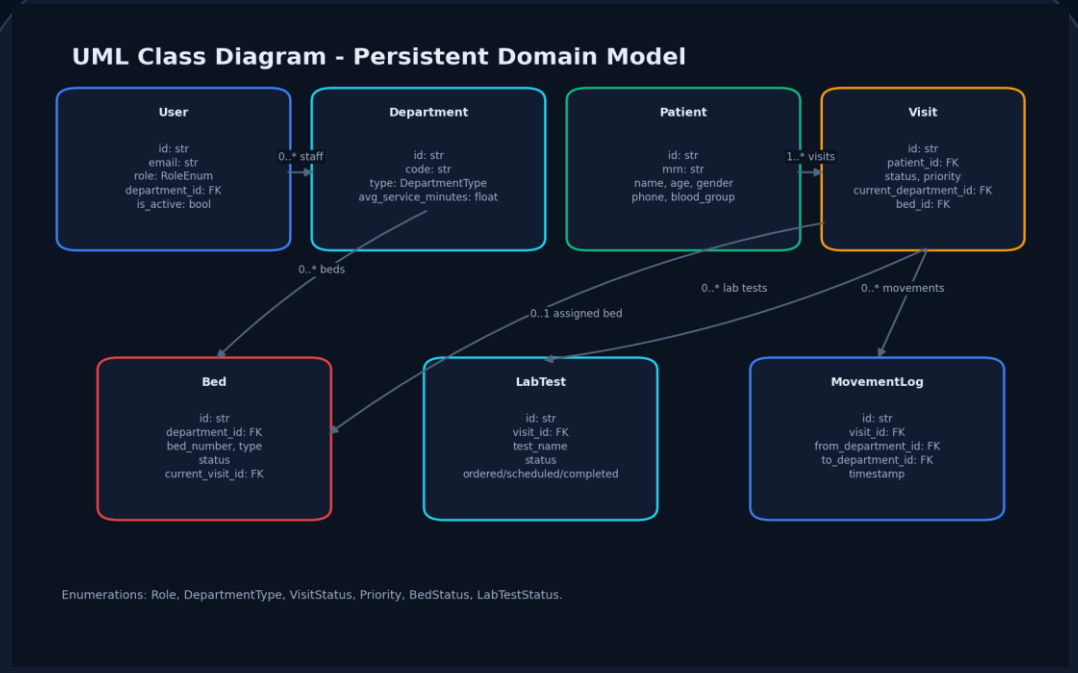


Level 2: register and prioritise

Database and Classes



Seven persistent entities



Domain model

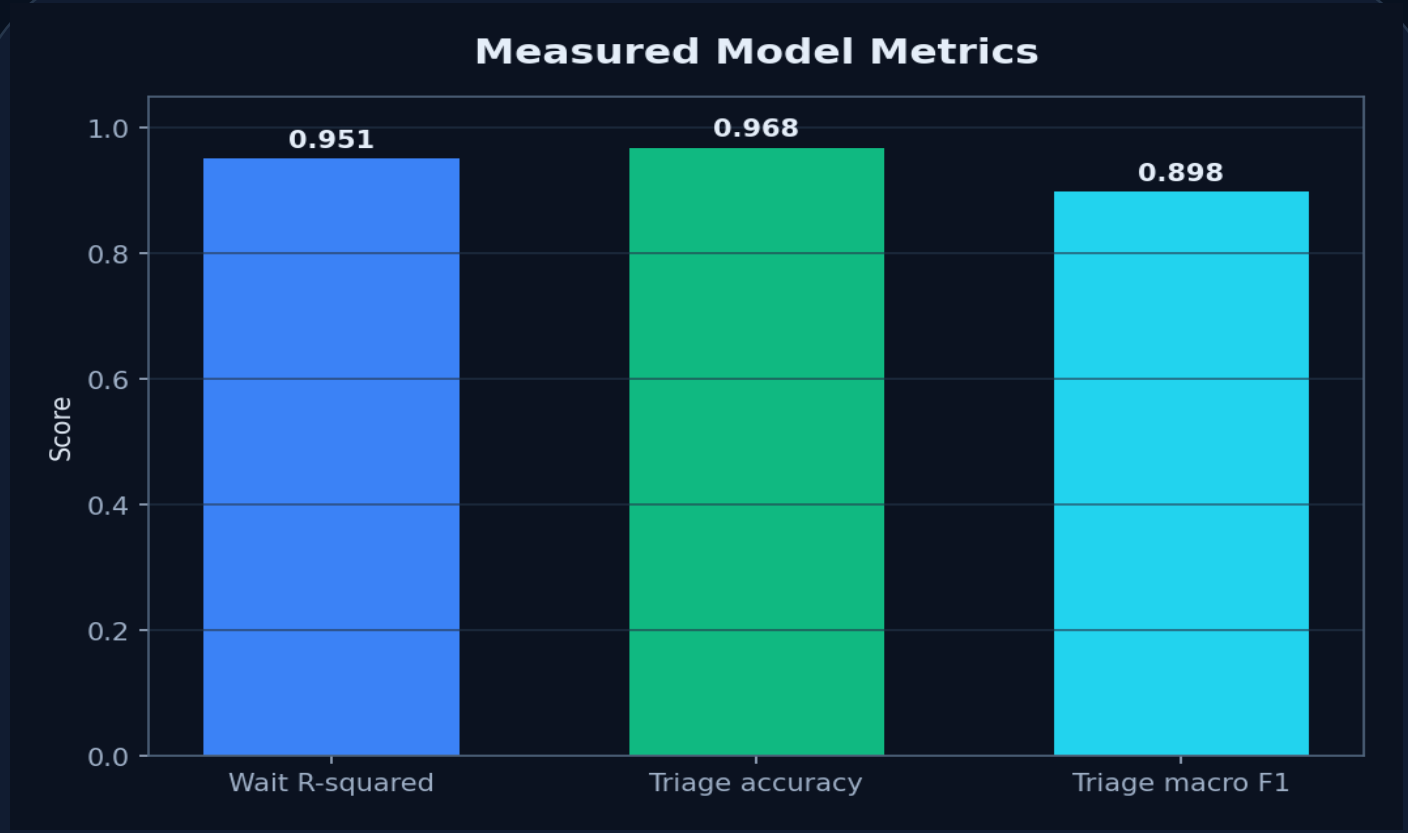
Implementation Highlights

- JWT login and role dependencies.
- Priority-first queue service and movement logs.
- Bed admission/release and laboratory-state routes.
- Public queue board and administrator KPIs.
- Axios token injection and WebSocket refresh.
- Joblib artefacts with stored encoders and feature order.

22 Application routes

7 Persistent entities

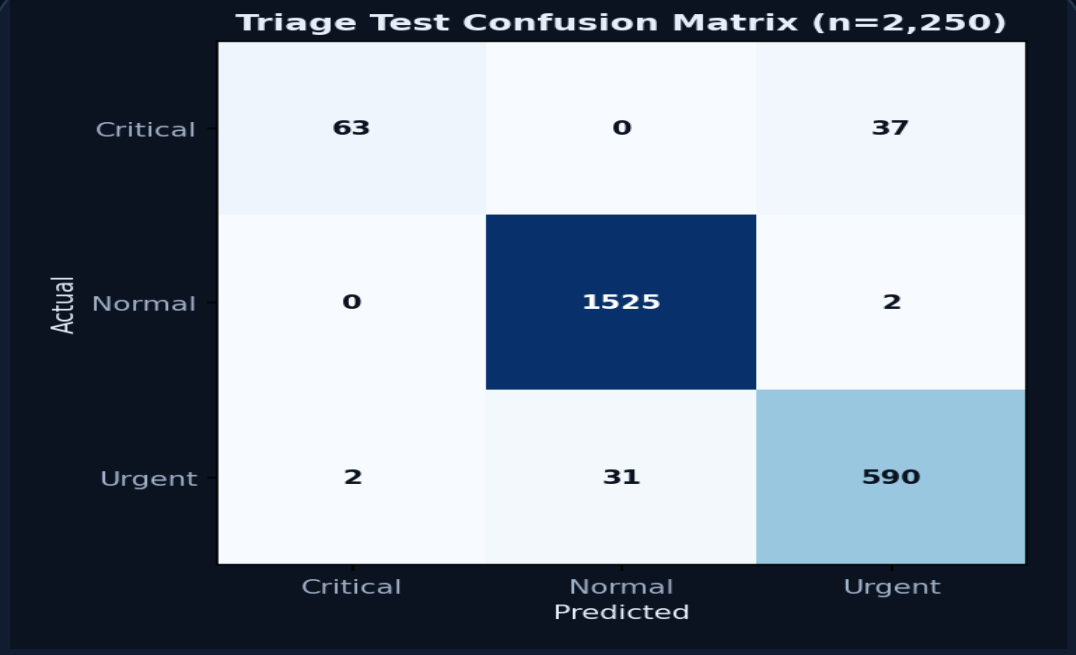
Model Performance



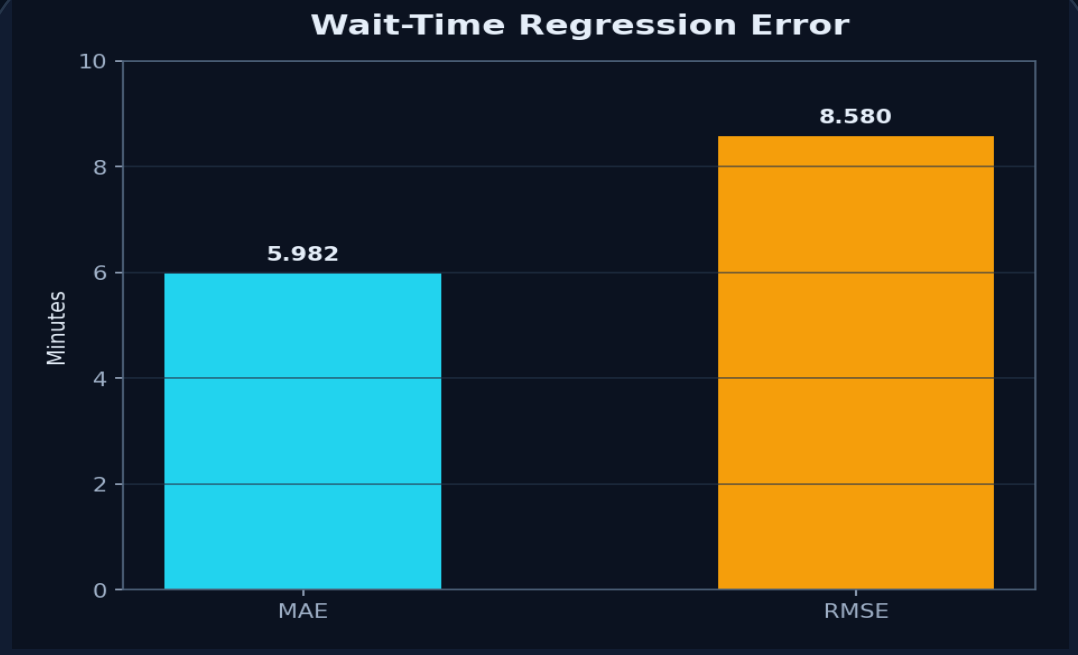
Generator-matched test scores

- Wait R^2 0.951041
- Triage accuracy 0.9680
- Macro F1 0.898475
- Not clinical validation

Confusion and Error



Critical recall = 63%



MAE 5.982 · RMSE 8.580 min

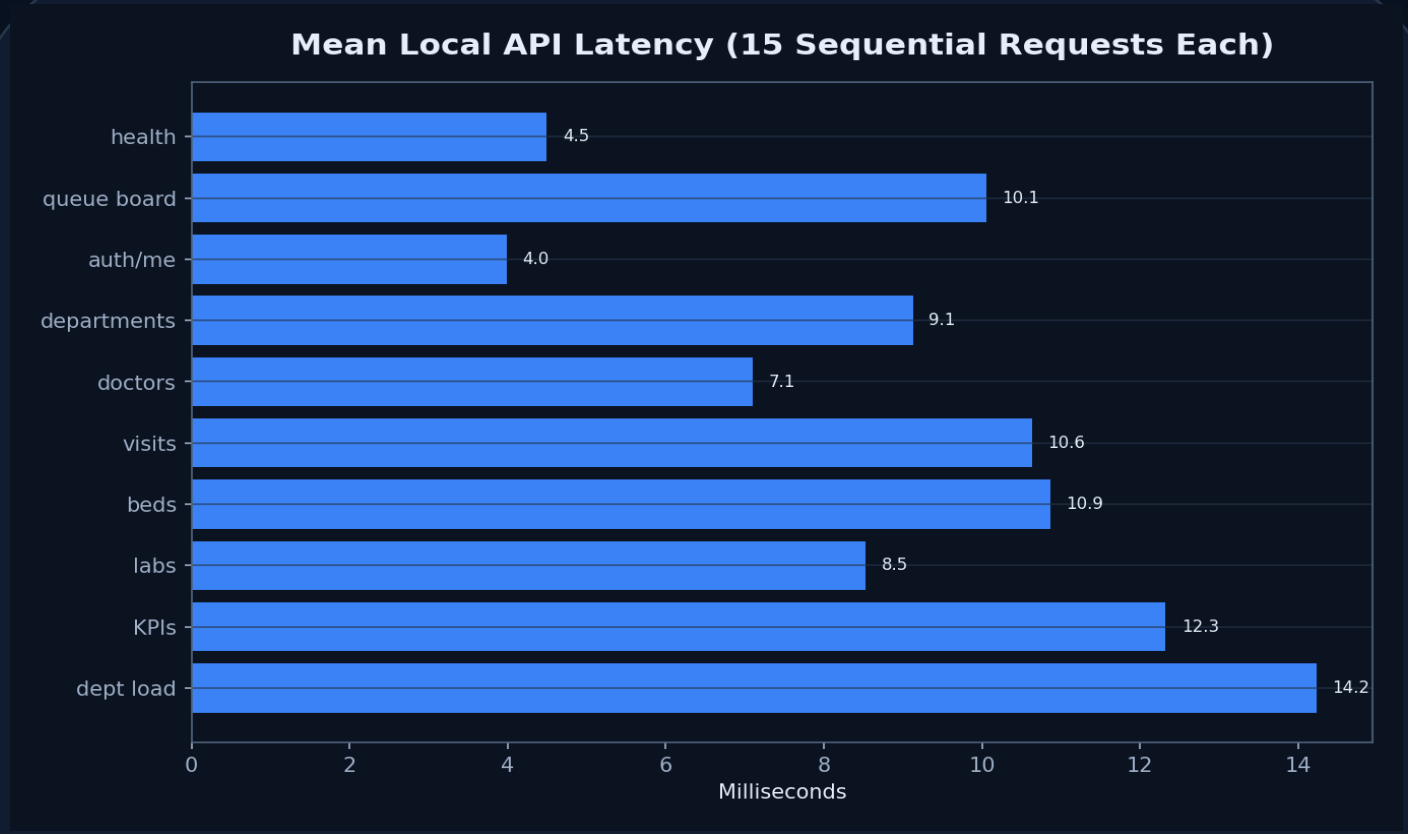
Parameter and Limitation Analysis

- Fixed Random Forest: 200 trees, max depth 10.
- Fixed Gradient Boosting: 150 estimators, max depth 3.
- No threshold sweep or calibration artefact exists—none is invented.
- Overall accuracy hides 37 critical-to-urgent errors.
- Real-data temporal and subgroup analysis remains future work.

63% Critical recall

None Threshold sweep

Performance and Testing



Sequential local mean latency

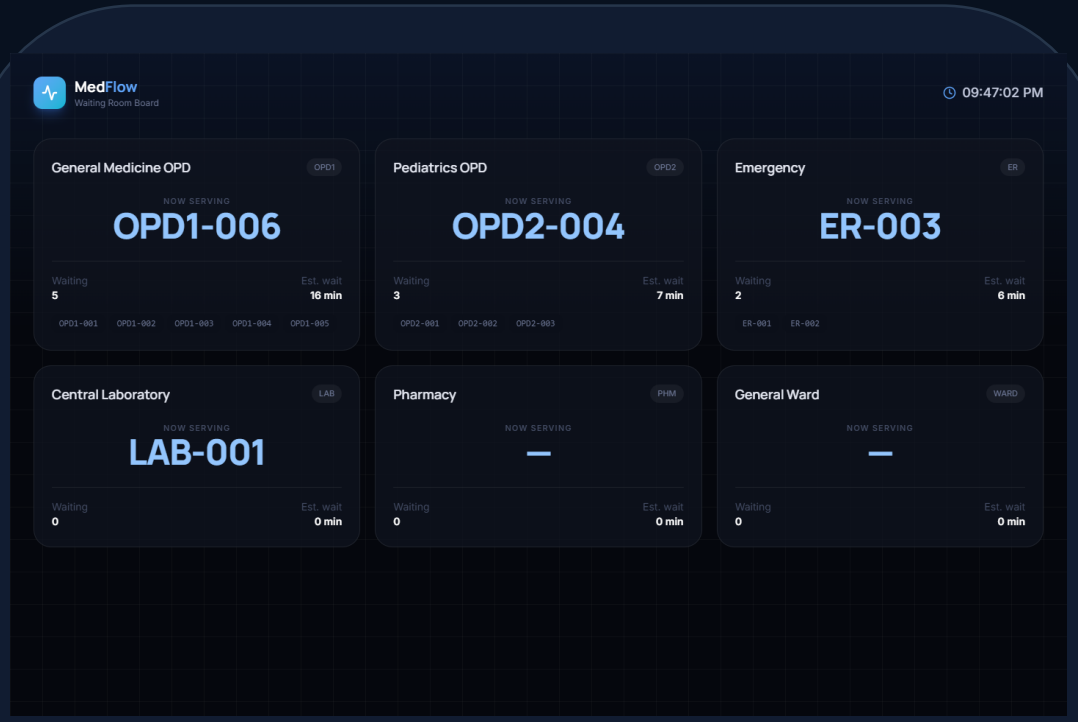
- 150/150 HTTP 200
- 3.985–14.231 ms means
- 46.657 ms WebSocket
- Build passed
- 14 lint warnings
- No load test

Comparison with Base Paper

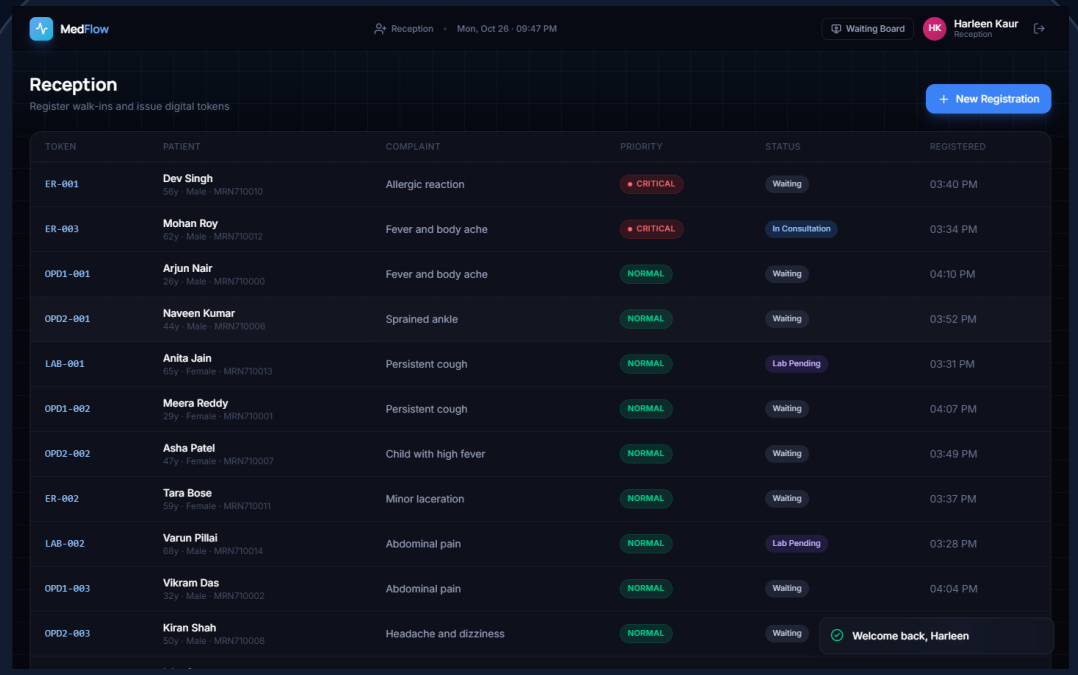
- Alleviate: patient access, specialist recommendation, prescriptions, maps and personalised plans.
- CareFlow: internal queue, visit, laboratory and capacity coordination.
- Different tasks, datasets, splits and conditions make numeric comparison invalid.
- The valid comparison is functional coverage and research gap.

Base	**Patient-facing**
Ours	**Operations-facing**

Output 1 · Public and Reception

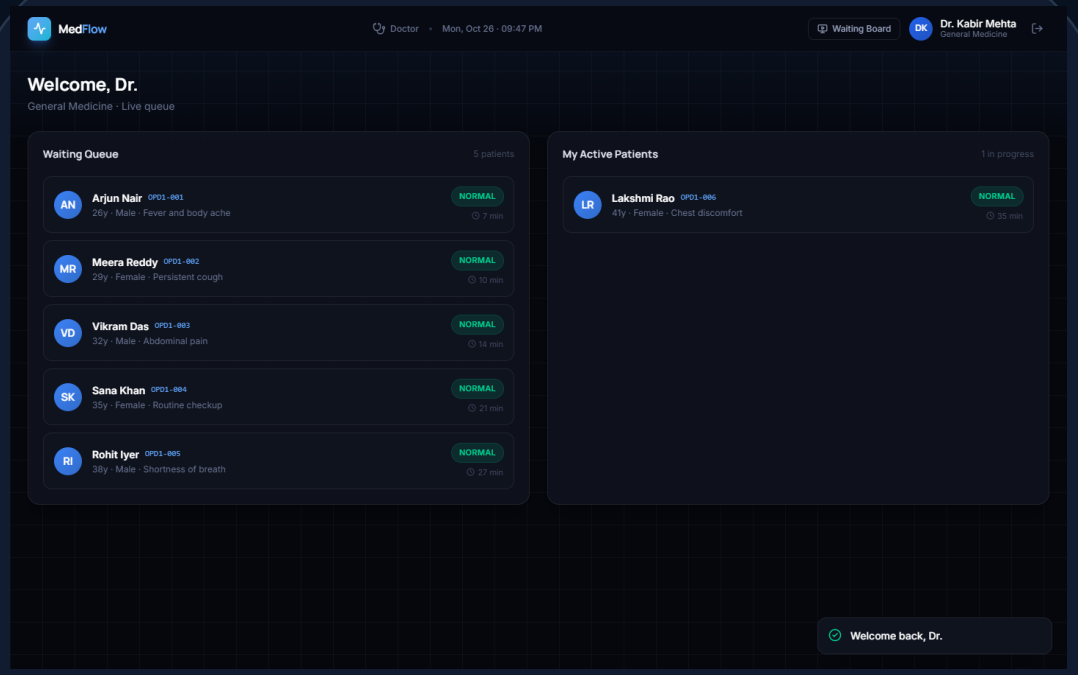


Public token board

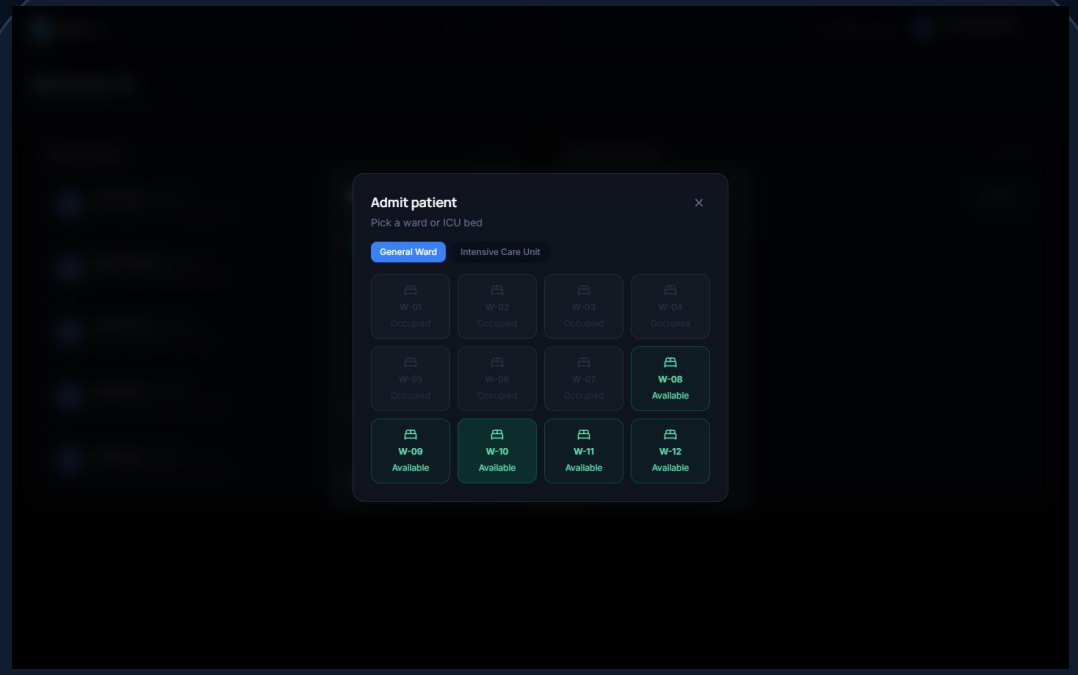


Registration list

Output 2 · Clinical Workflow

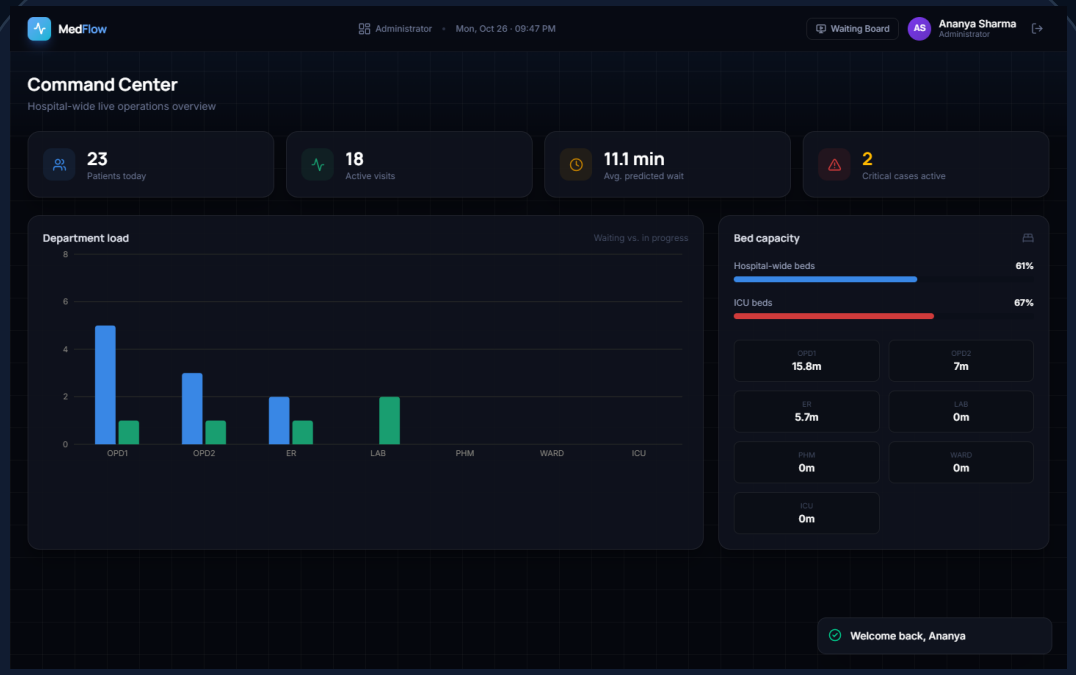


Doctor queue

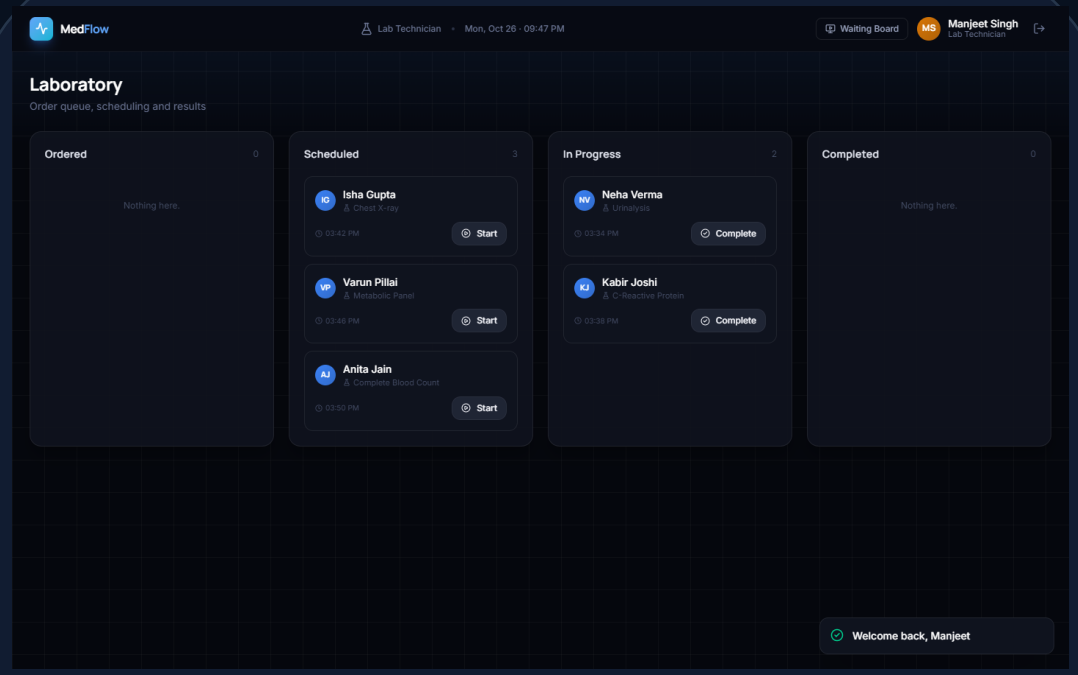


Available-bed selection

Output 3 · Operations



Administrator overview



Laboratory work queue

Conclusion

- Implemented one coherent multi-role patient-flow prototype.
- All 150 measured read requests returned HTTP 200.
- Models produced reproducible synthetic-test results.
- Critical recall and missing clinical validation define the safety boundary.
- No real wait-reduction or deployment claim is made.

Achieved Functional integration

Limited Clinical evidence

Future Scope

- Governed real-data collection and temporal validation.
- Cost-sensitive triage, calibration and uncertainty display.
- Automated state-transition and end-to-end tests.
- HL7 FHIR integration with EHR, lab and admission systems.
- PostgreSQL, message broker, authenticated live channels and observability.
- Clinician-led usability, safety and prospective impact study.

1 **Validate safety**

2 **Scale securely**

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THANK YOU

Questions and suggestions are welcome